

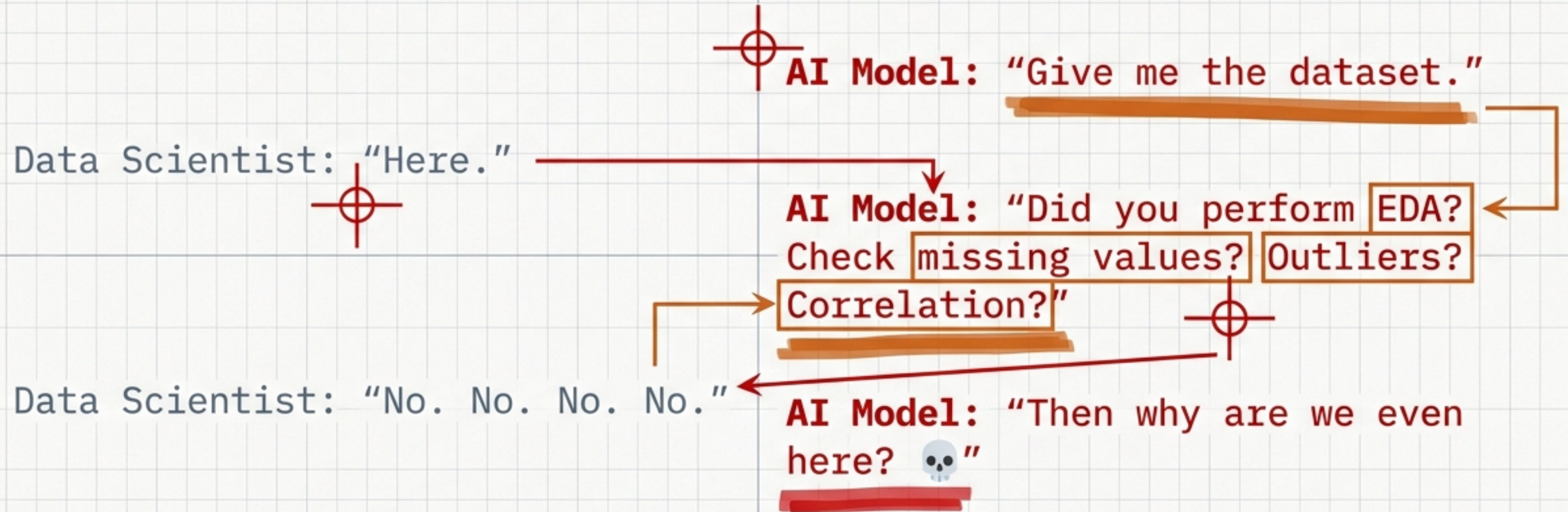
Exploratory Data Analysis

Before asking a machine to learn from data, learn something about the data yourself.

256	1024	-999	0.001	0.001	15.7	ERROR	NaN	NaN	0.000	NaN
1024	254	-999	0.001	NaN	15.7	2.7	NaN	NaN	7.99	NaN
1024	1024	-999	0.001	100.7	NaN	VALID	NaN	NaN	NaN	ERROR
1024	-999	-999	0.001	256	VALID	VALID	COMPLETE	NaN	NaN	UNKNOWN
NULL	NULL	NULL	PENDING	ERROR	NaN	VALID	NULL	NULL	NULL	COMPLETE
1024	1024	-999	0.001	ERROR	NaN	VALID	NULL	NULL	NaN	UNKNOWN
1024	1024	256	0.001	15.7	NaN	256	NaN		30	UNKNOWN

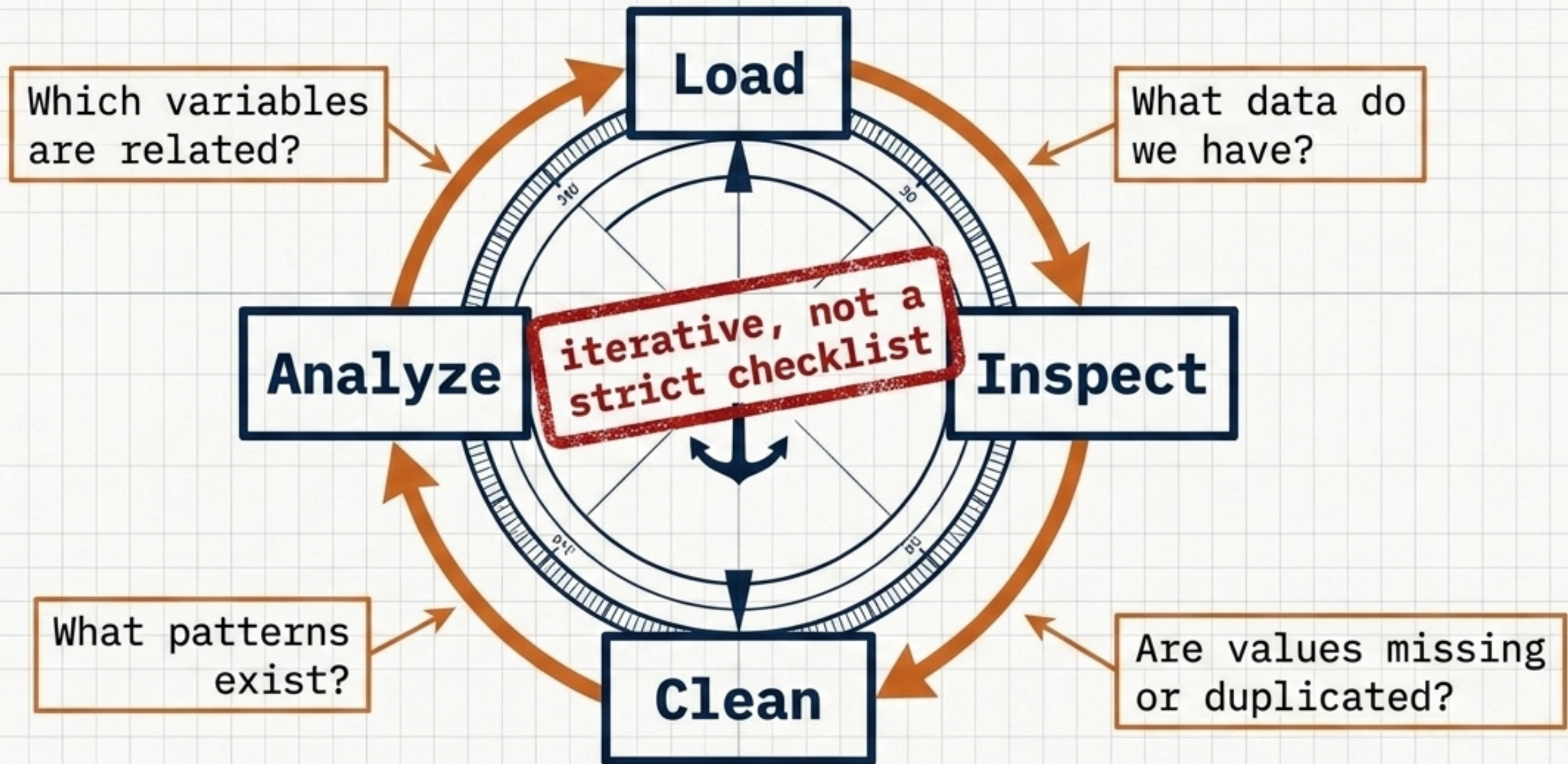
Train model first, understand data later.
– Famous last words 💀

The Prerequisite for Intelligence



EDA = Getting to know your dataset. It is the active process of identifying problems before they become modeling problems.

The Interrogation Loop



Anatomy of a Dataset

The first macro-look at the structure. Now we know what we are dealing with.

passenger_id	pclass	name	sex	age	sibsp	parch	ticket	fare	cabin	embarked
1	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
2	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
3	3	Rockmont, Mr. Calen Harris	male	25.0	1	0	A/4 21150	7.9700	NaN	S
4	4	Brokson, Mr. Walton	male	27.0	0	1	A/5 21271	2.2000	NaN	S
5	4	Braund, Mr.	male	15.0	1	0	A/5 10392	7.9500	NaN	S
6	5	Braund, Mr.	male	23.0	0	0	A/6 20345	7.2500	NaN	S
7	3	Braund, Mr. Harris	male	22.0	1	0	A/5 20393	4.4000	NaN	S

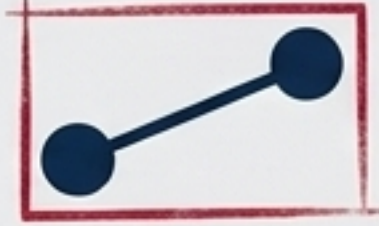
Feature meanings
& Data types?

Are values missing?
Non-null counts?

How much data
do we have?
Memory usage?

The Tools:
`df.head()`, `df.info()`, `df.shape`

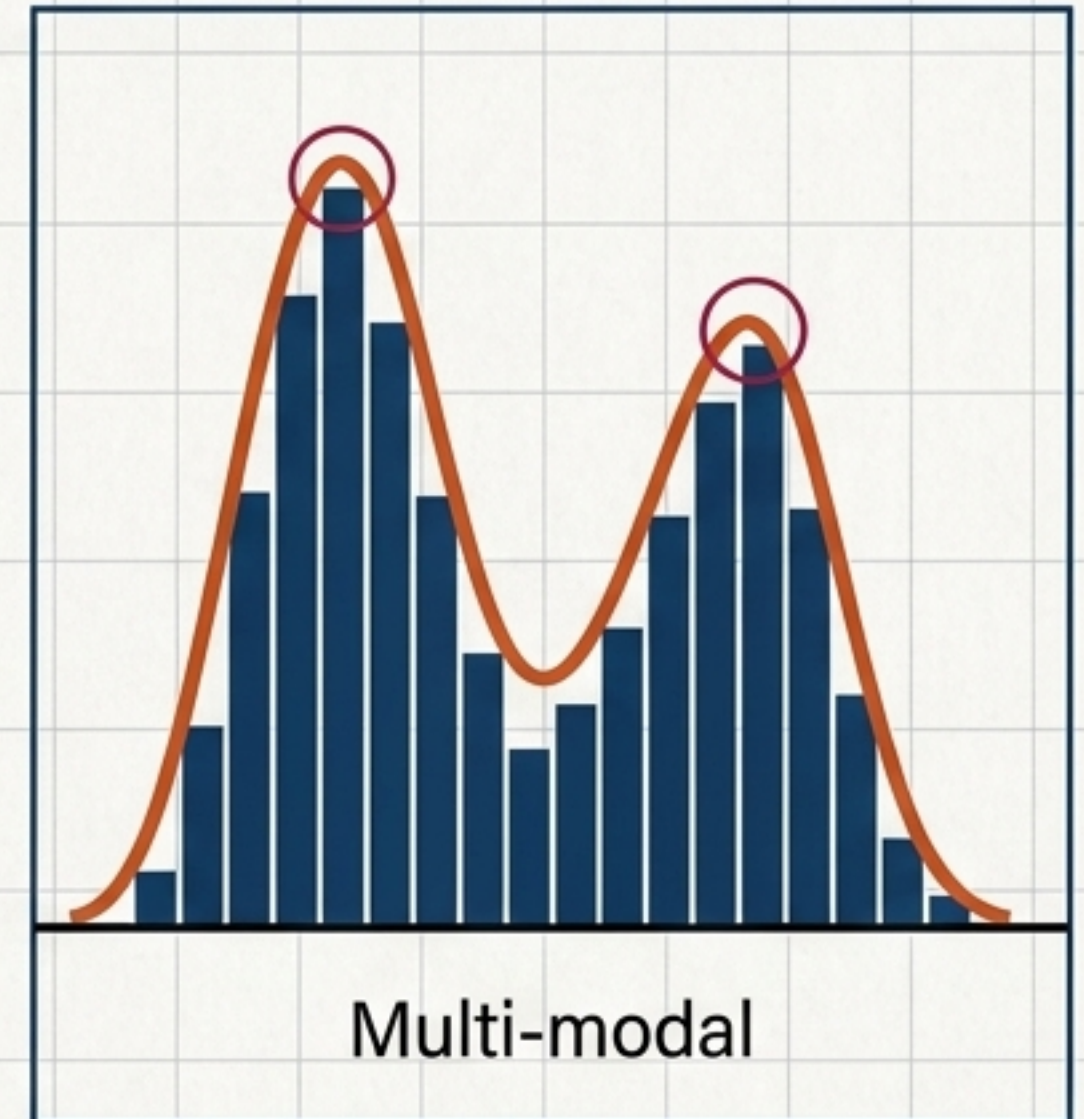
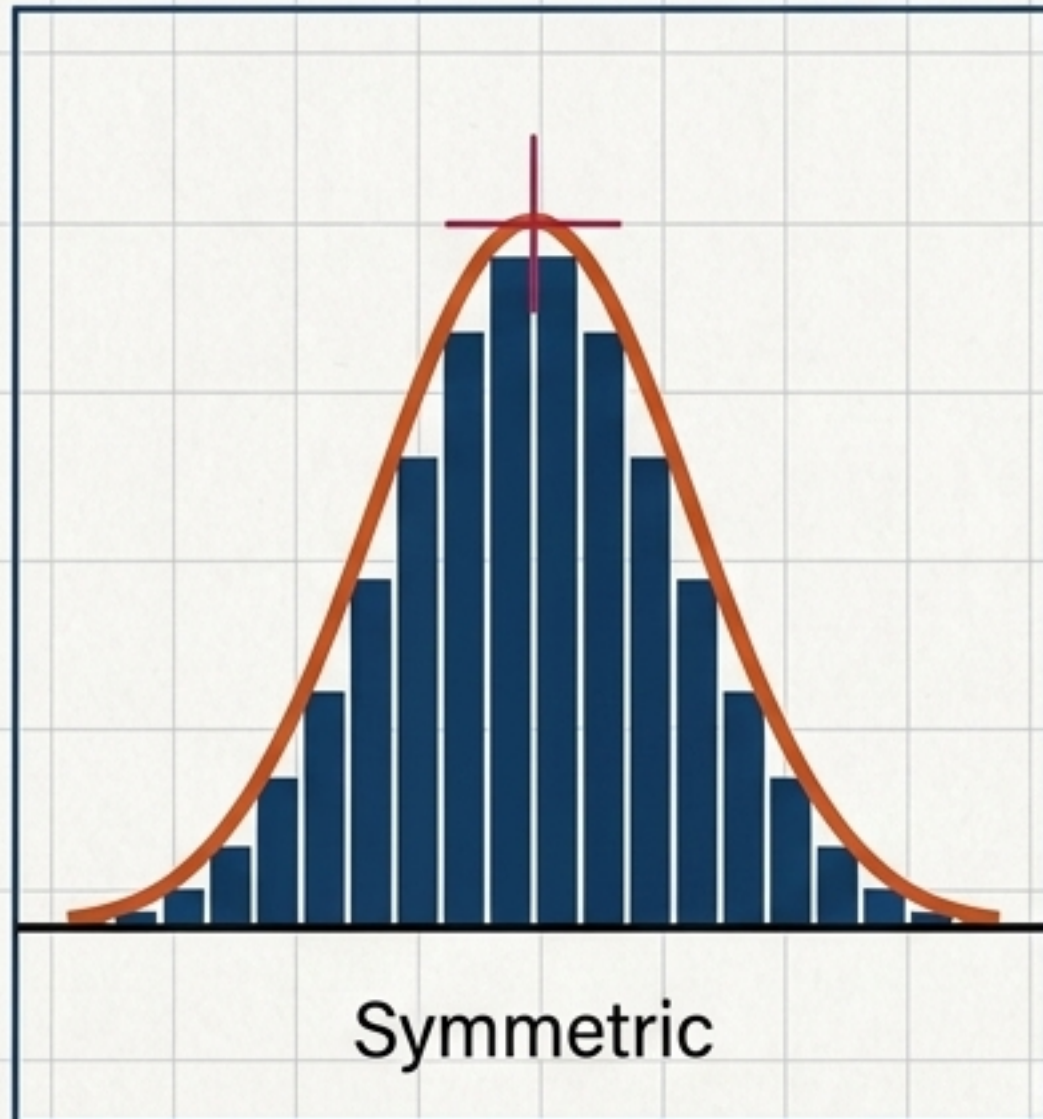
The Dimensionality Matrix

		
Univariate Analysis	<u>Bivariate Analysis</u>	Multivariate Analysis
<u>Variables:</u> One Variable	<u>Variables:</u> Two Variables	<u>Variables:</u> Multiple Variables
Goal: Understand distributions & extreme values.	Goal: Study relationships.	Goal: Understand complex interactions.
Key Questions: Average? Range? Unusual values?	Key Questions: Does salary increase with age?	Key Questions: How do several variables interact simultaneously?
Tools: Histograms, <code>df.describe()</code>	Tools: Scatter plots.	Tools: <u>Correlation matrices</u> (Crucial for Machine Learning).

Decoding Numerical Distributions

A distribution describes how values are spread. Are they spread out? Concentrated?

Diagnostic Rule: Understanding distributions dictates our preprocessing and modeling approaches.

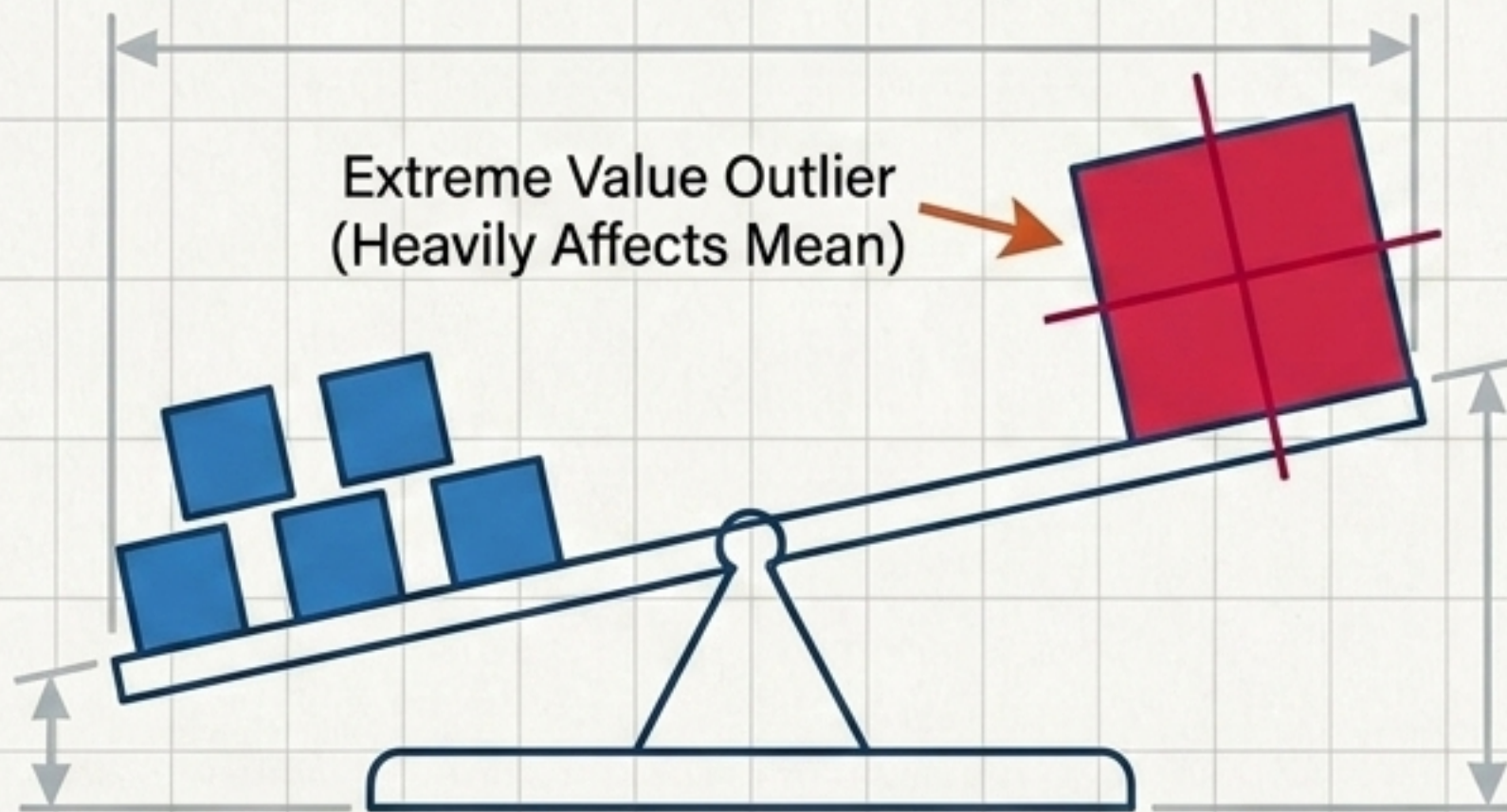


The Center of Gravity: Mean vs. Median

Mean and median tell different stories. The extreme value heavily affects the mean.

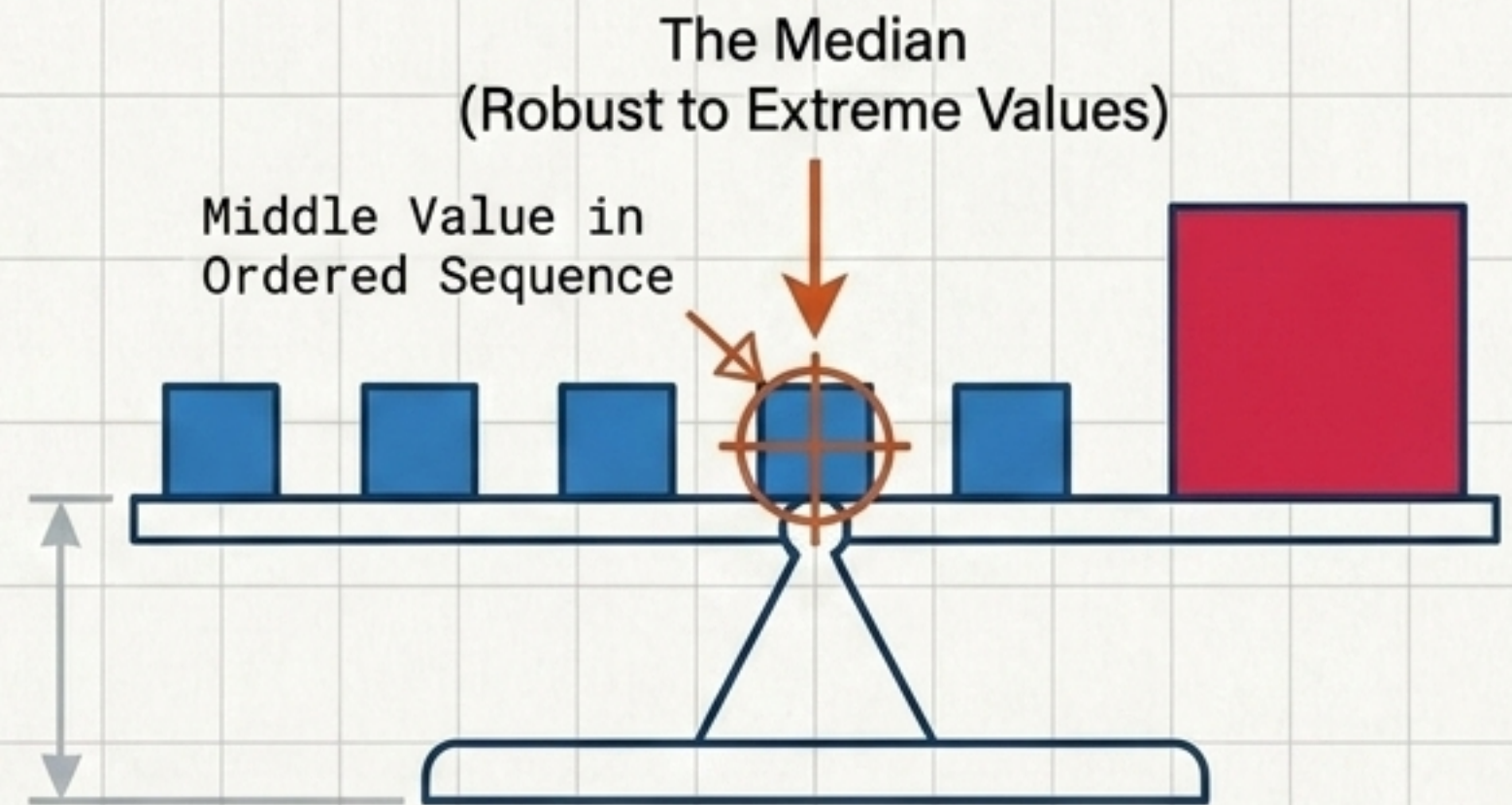
The median is robust to extreme values.

The Mean



The Mean

The Median



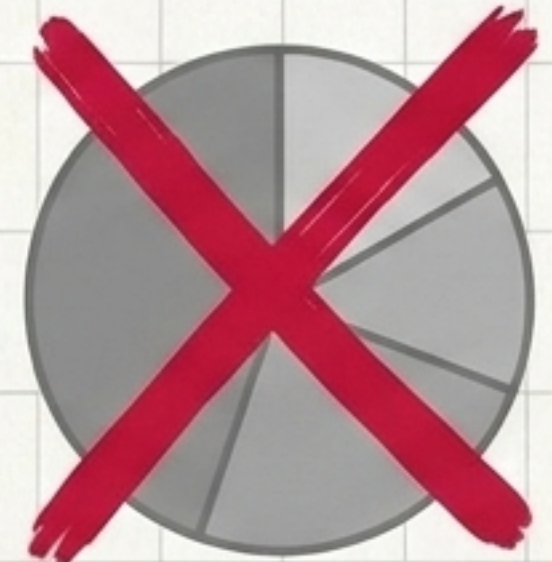
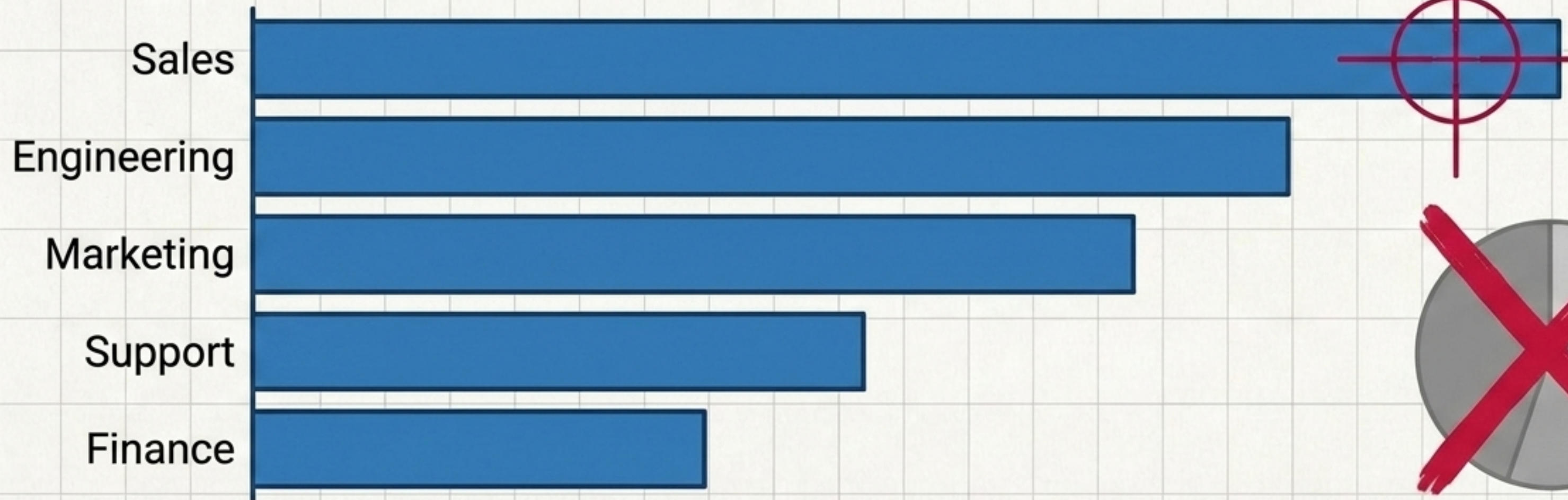
The Median

Non-Numerical Evidence

Categorical Analysis studies frequency and category distribution.

Example: Grouping by 'Department' or 'Job Title'.

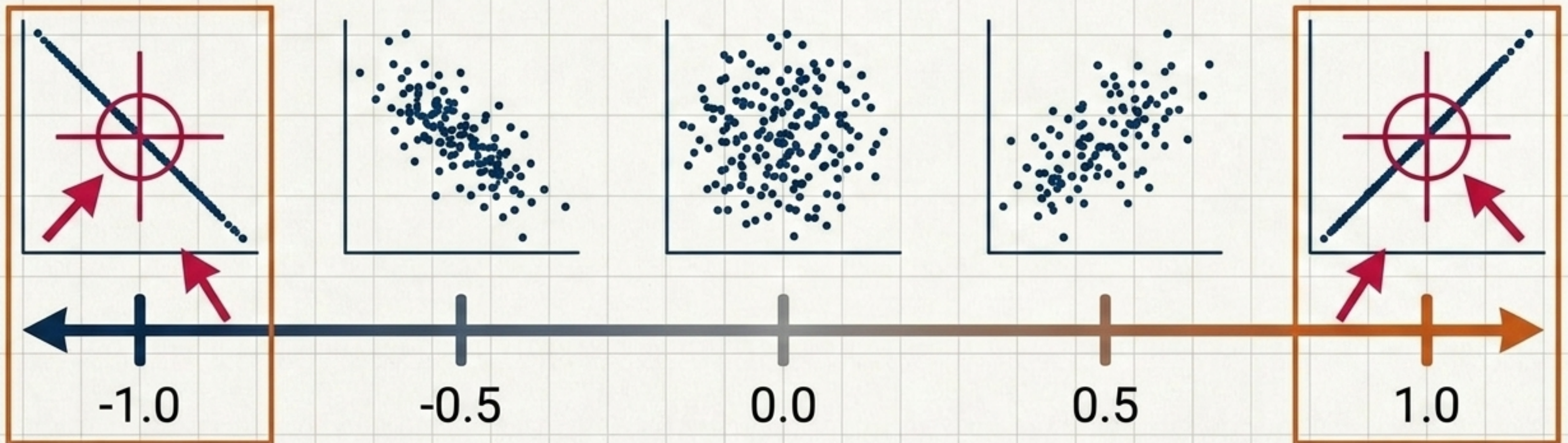
Tool: `df['Category'].value_counts()`



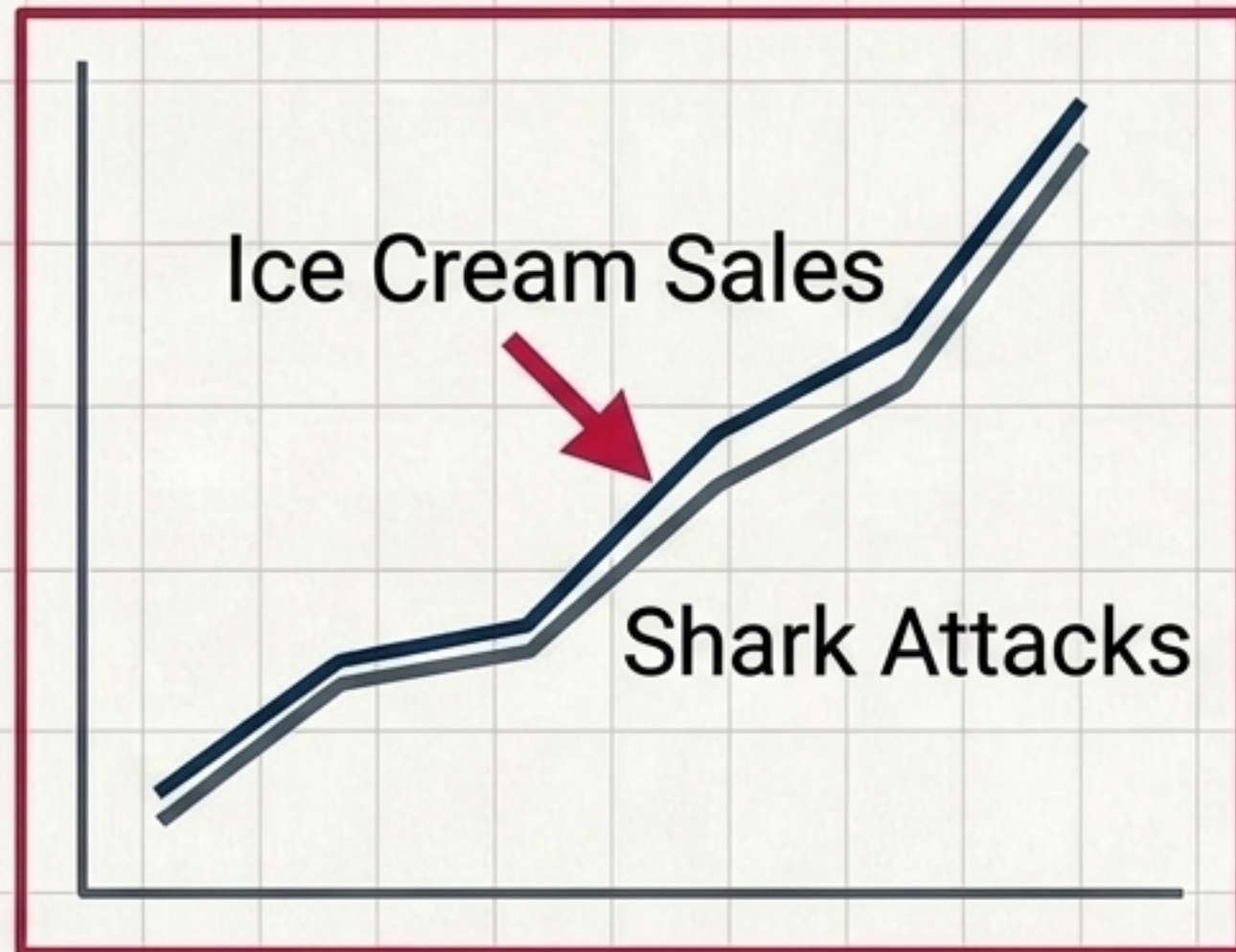
The Correlation Spectrum

Correlation measures the strength and direction of a linear relationship between numerical variables.

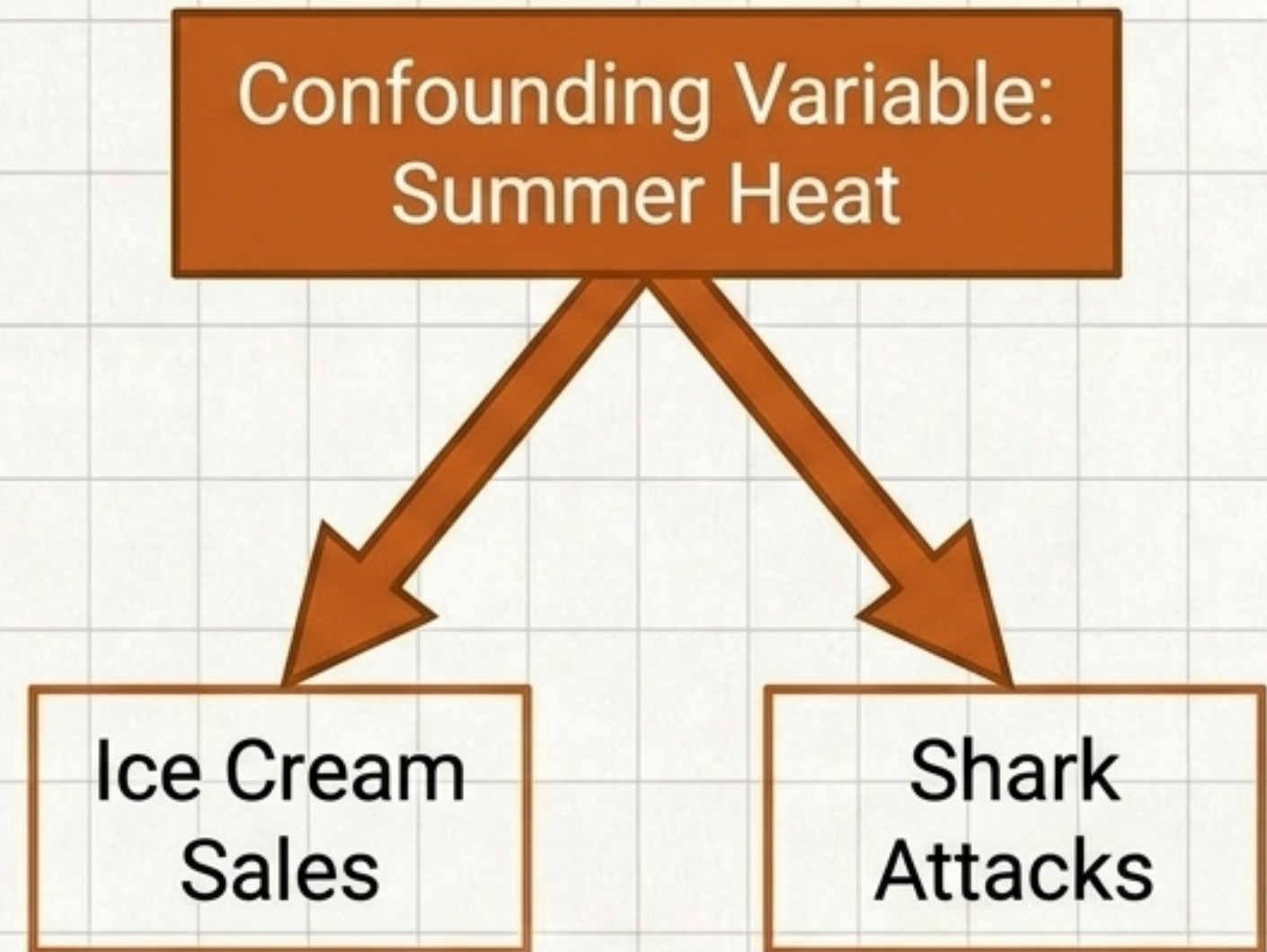
Multiple features? Use a Correlation Matrix to spot strong relationships or redundant features.



The Causation Trap



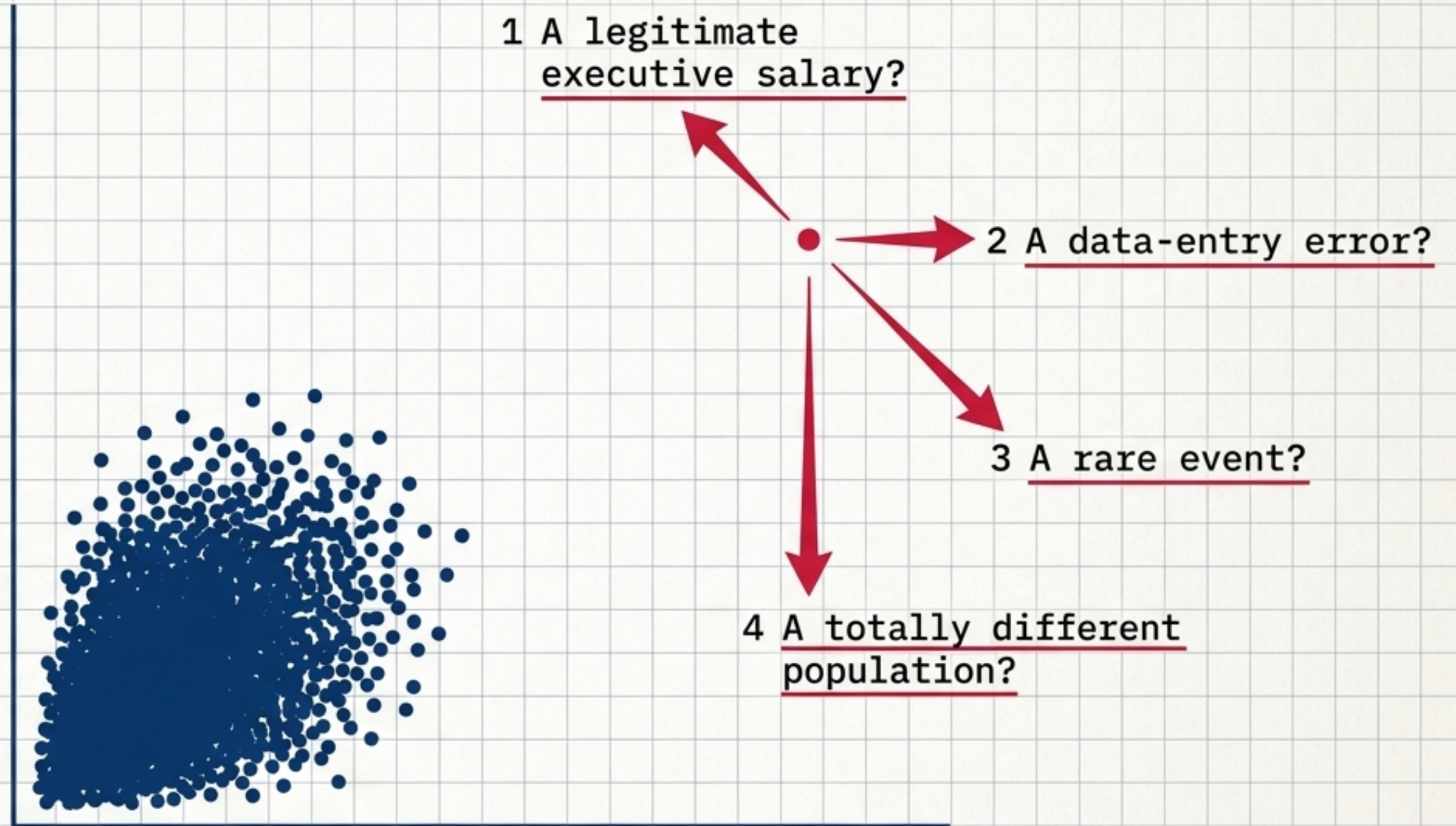
Highly Correlated.



Correlation does not imply causation.

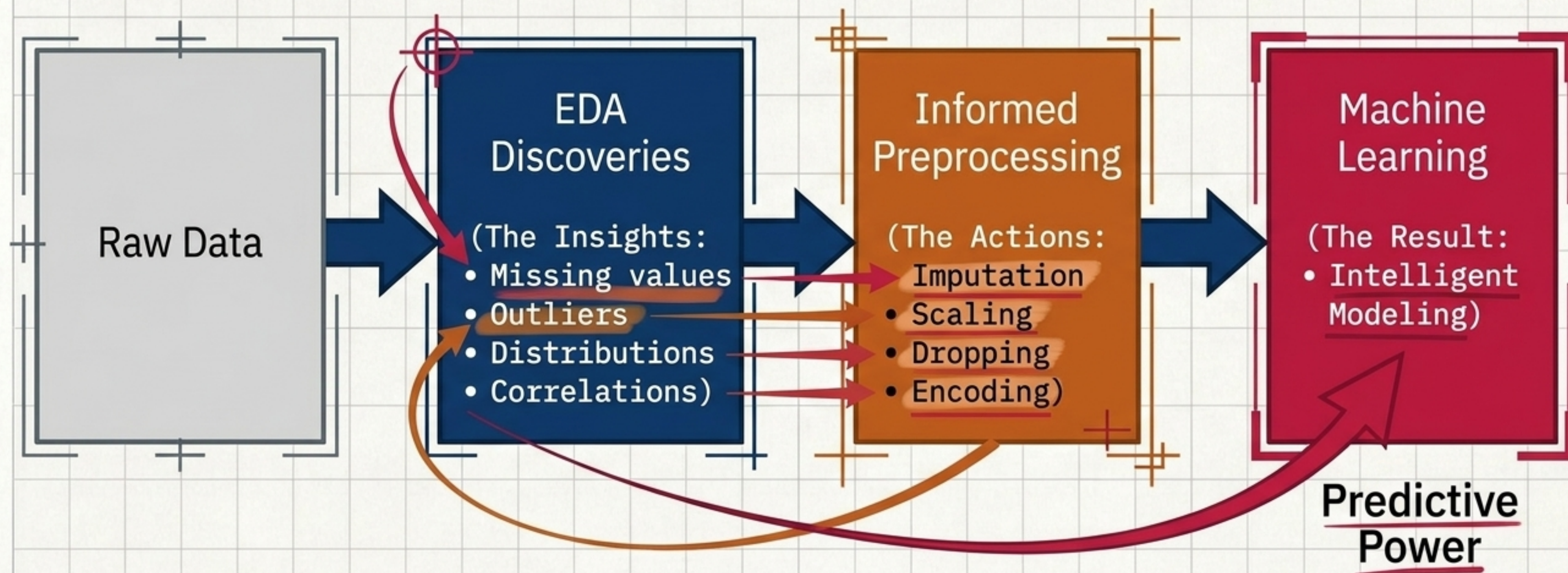
Anatomy of an Outlier

An outlier isn't automatically an error. It is a question. Always investigate.



The Bridge to Machine Learning

EDA doesn't replace preprocessing. It tells you exactly how to preprocess.



The Investigator's Challenge

The core mindset is not: **"Which model should I use?"**

It is: **"What is actually happening in my data?"**

Mini Project Challenge:

1. Take any CSV dataset.
 2. Detect missing values & duplicates.
 3. Map numerical & categorical distributions (Histograms/Bar charts).
 4. Plot relationships & correlations (Scatter plots).
 5. Write 5 observations. Don't just generate charts.
- Extract insights. ←

The Complete Python Data Journey concludes here. Next: Machine Learning.